

DOCUMENTATION TECHNIQUE

Documentation Technique & Architecture du Pipeline

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CLI To-Do List Manager - Technical Specification & Architecture Document

1. Executive Summary & Architecture Overview

1.1 Executive Brief

A command-line interface (CLI) application designed to manage a task list persisted via a local JSON file. The system provides core CRUD operations (add, list, complete, remove, clear) with a specific focus on data integrity, input validation, and dual-mode output (human-readable and standard JSON).

1.2 Maturity Assessment

The project provides a clear functional contract for CLI commands, but is currently in REFINEMENT status. While completeness is high, the lack of a defined JSON schema for the storage file and the absence of a formal testing suite for edge cases (specifically for malformed JSON handling) represent significant structural gaps that must be addressed before implementation.

1.3 Technical Stack

- **Storage Format:** JSON
- **Persistence Layer:** Flat-file system (`~/ . todos . json`)

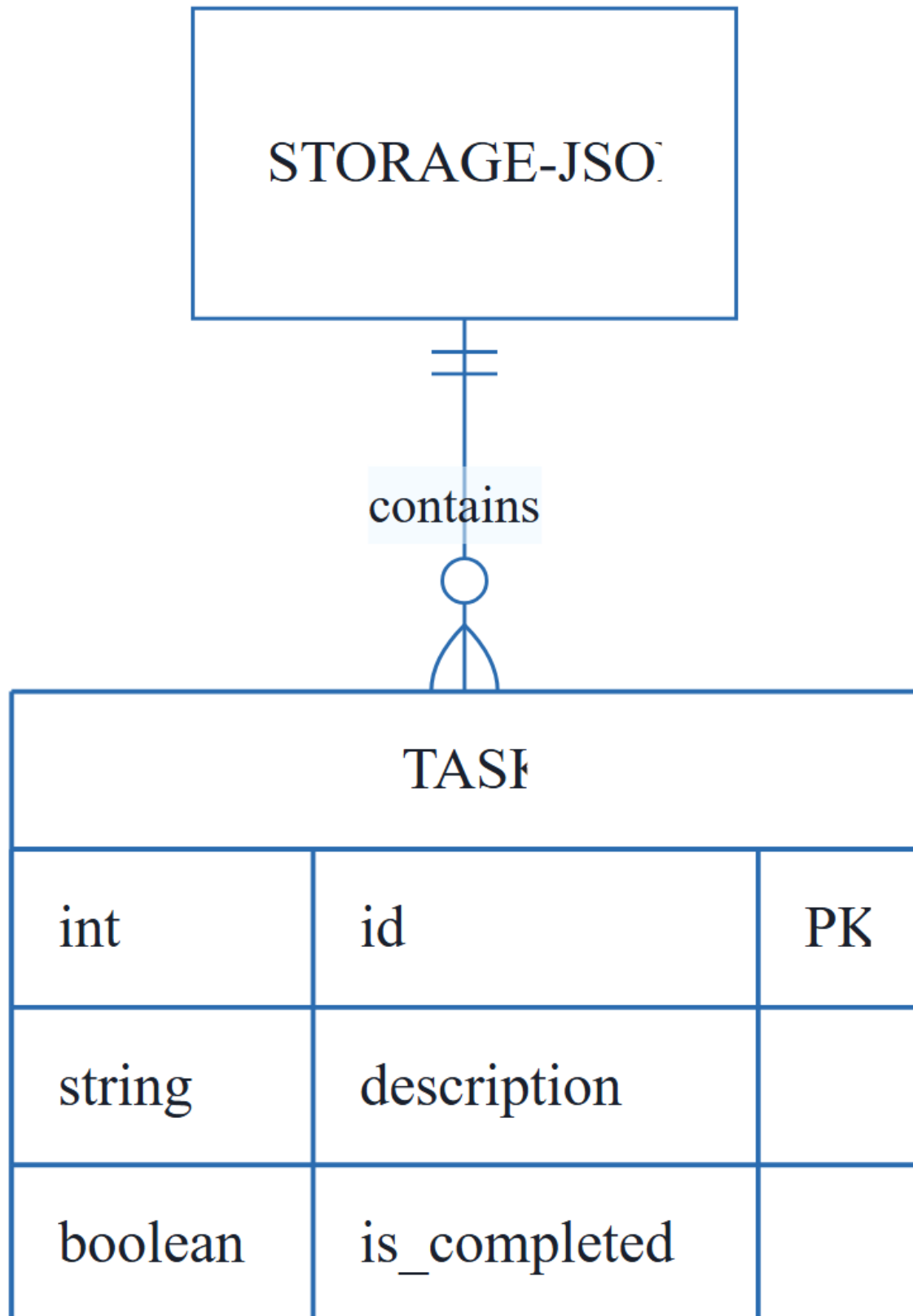
1.4 Architectural Constraints

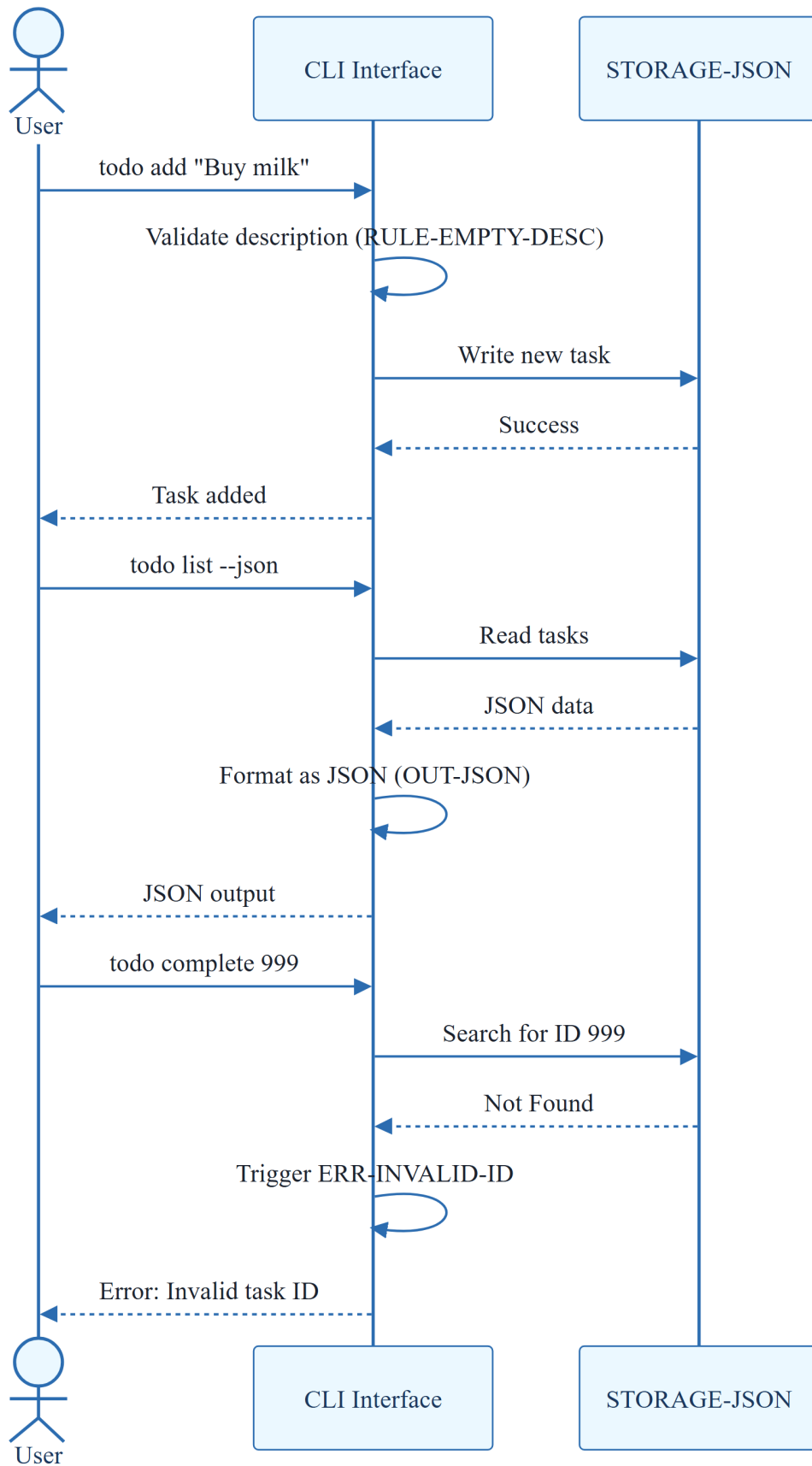
- Descriptions consisting only of whitespace or empty strings must be rejected prior to file system write operations.
- Malformed storage files must trigger a user-friendly error message and return a non-zero exit code.
- Invalid task IDs must trigger a clear, user-friendly error message.
- JSON output mode must be standard-compliant and accurately reflect the current state of the storage file.
- Human-readable output must explicitly include Task ID, description, and completion status.

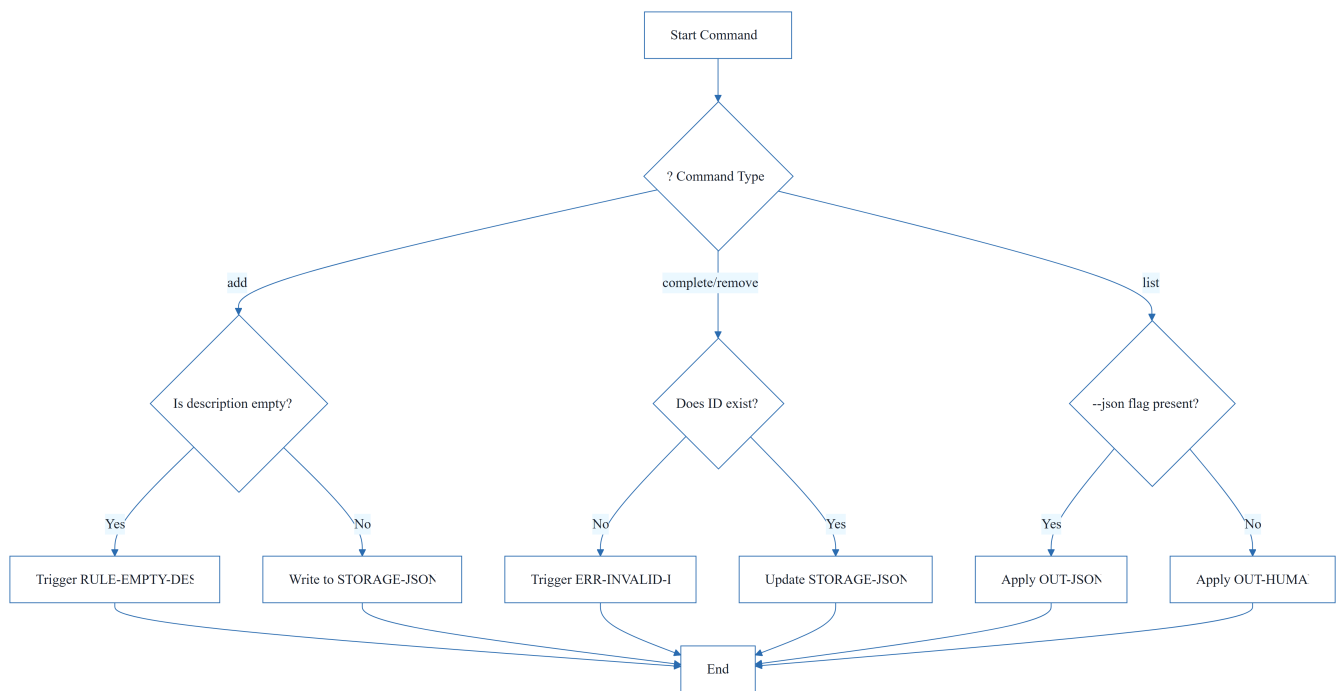
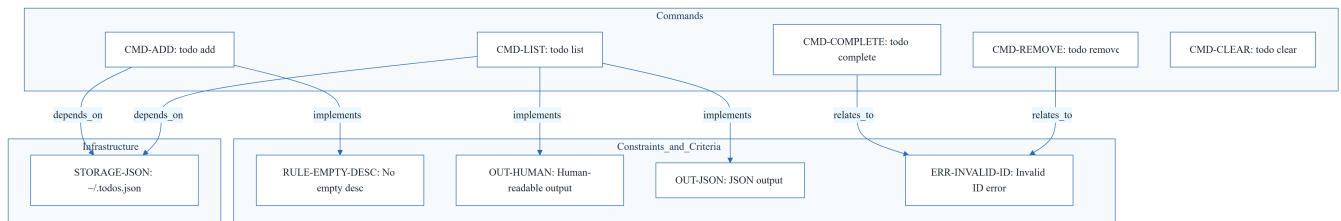
1.5 Critical Dependencies

- File system access to `~/ . todos . json` for state persistence.
- Strict dependency between `CMD-ADD/CMD-LIST` and the JSON storage backend.
- Logical coupling between `CMD-COMPLETE/CMD-REMOVE` and the ID validation mechanism.
- Standard JSON parser for output and storage synchronization.

2. Architecture Workflows

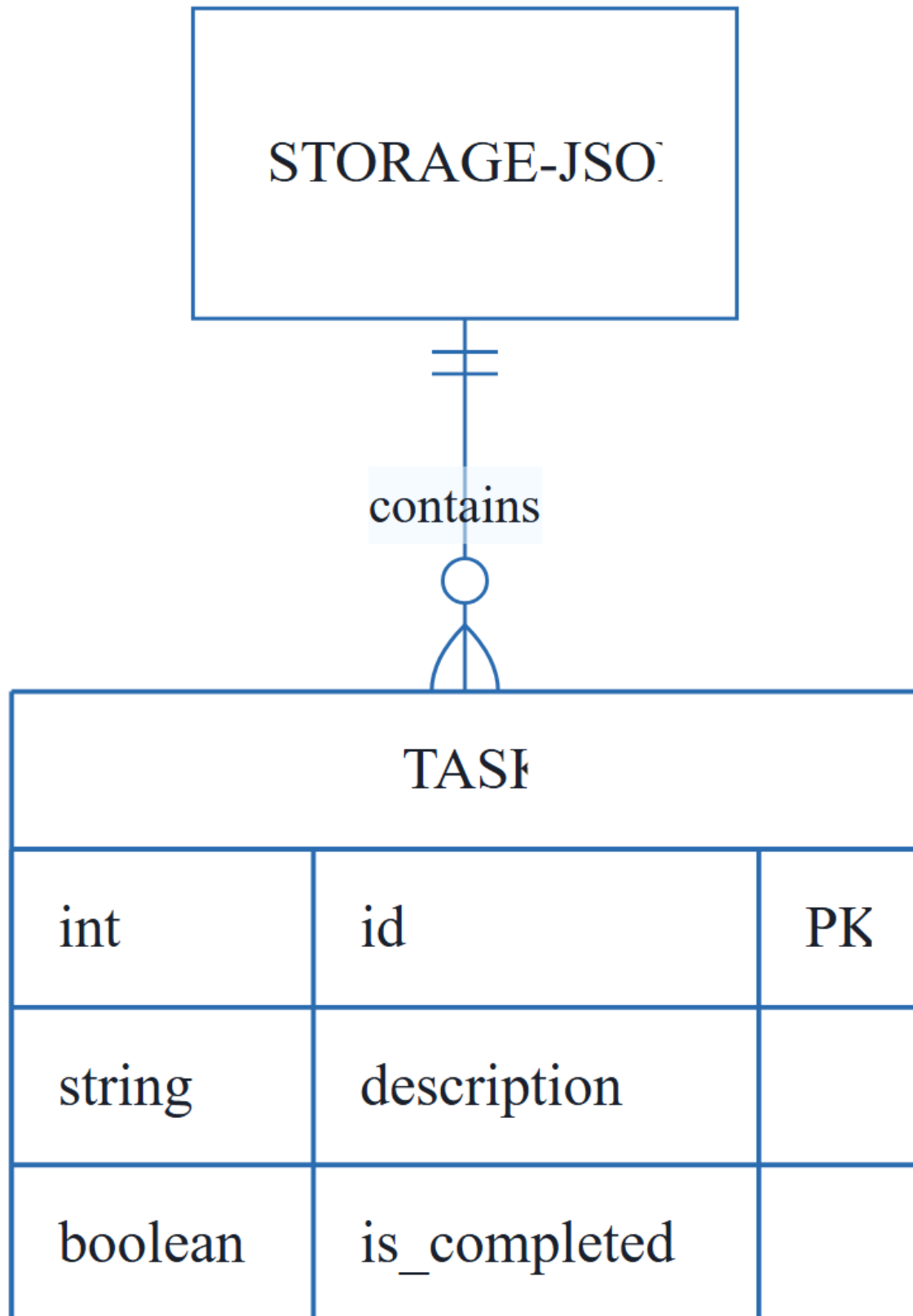




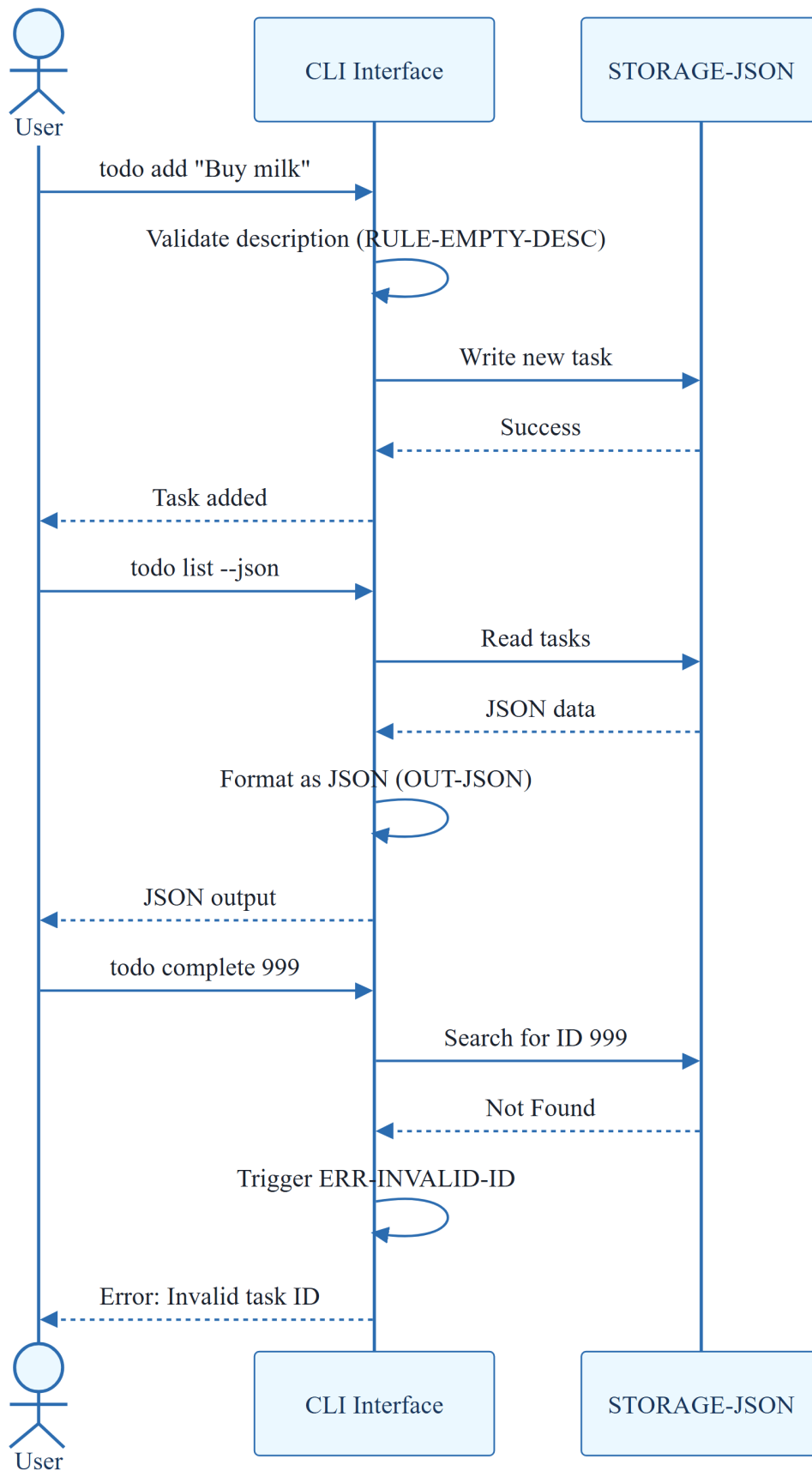


& Visual Diagrams

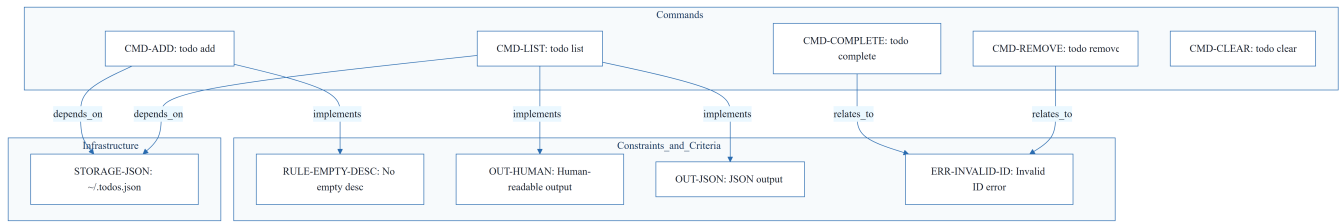
2.1 CLI To-Do Manager Data Model



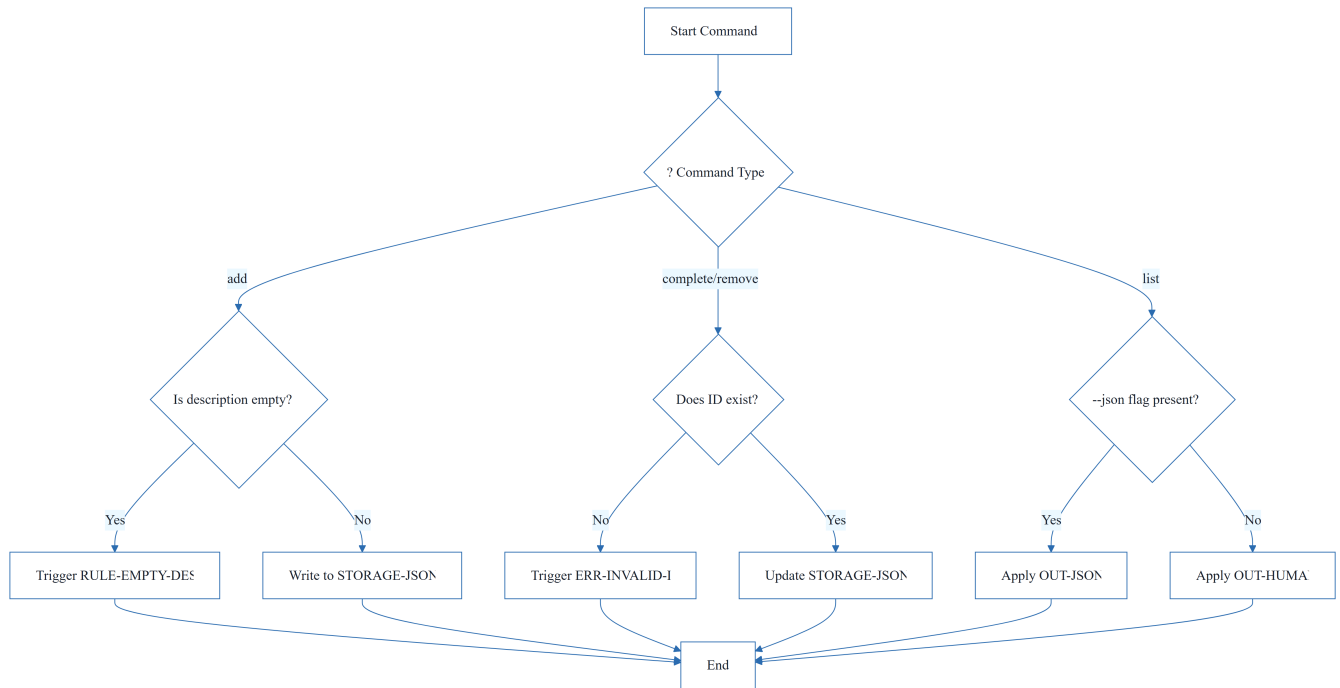
2.2 Command Execution Sequence



2.3 Requirements Traceability Matrix



2.4 Command Logic Workflow



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Type	Requirement / Rule Description	Source Section
CMD-ADD	Task	Implement todo add command	todo add
RULE-EMPTY-DESC	Constraint	Reject empty or whitespace-only descriptions before file write	todo add
STORAGE-JSON	Dependency	Storage file at ~/.todos.json	todo add
CMD-LIST	Task	Implement todo list [--json] command	todo list [--json]

Identifier	Type	Requirement / Rule Description	Source Section
OUT - HUMAN	Acceptance Criterion	Human-readable output must show ID, description, and status	Output Rules
OUT - JSON	Acceptance Criterion	JSON output must be standard-compliant and reflect current list	Output Rules
CMD - COMPLETE	Task	Implement <code>todo complete</code> command	<code>todo complete</code>
CMD - REMOVE	Task	Implement <code>todo remove</code> command	<code>todo remove</code>
CMD - CLEAR	Task	Implement <code>todo clear</code> command	<code>todo clear</code>
ERR - INVALID - ID	Constraint	Invalid IDs must produce clear user-friendly error messages	Error Rules
ERR - MALFORMED - FILE	Constraint	Malformed storage files must result in error message and non-zero exit code	Error Rules

3.2 Security Rules

- **Input Validation:** All inputs for `CMD-ADD` must be sanitized to prevent empty or whitespace-only entries from being persisted.
- **Error Handling:** System must ensure that malformed storage files do not cause application crashes but instead trigger a controlled exit with a non-zero code.

3.3 Data Models

The system utilizes a flat-file JSON database located at `~/ . todos . json`.

- **Entity:** Task
- `id`: Unique identifier (Integer/String).
- `description`: Textual content of the task.
- `is_completed`: Boolean flag indicating completion status.

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Gap	Priority	Remediation Advice
Dependencies & Integration Points	LOW	Specify if external libraries (like 'clap' for Rust or 'commander' for JS) are required.
Testing & Validation	HIGH	Define a test suite for each command, including edge cases for malformed JSON.
Implementation Notes & References	MEDIUM	Add details on the expected JSON schema for <code>~/ .todos.json</code> .
Open Questions & Uncertainties	LOW	Clarify if the ID is an integer, UUID, or user-defined string.

4.2 Remediation & Workflow

1. **Schema Definition:** Define the exact JSON structure to resolve Open Questions.
2. **Test Suite Design:** Create a matrix of test cases covering `ERR-INVALID-ID` and `ERR-MALFORMED-FILE`.
3. **Library Selection:** Select the CLI parsing framework based on the chosen language.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
ID	BUSINESS_DOMAIN	ERR-INVALID-ID	The unique alphanumeric reference used to target a specific entry for completion or deletion within the persistent storage.
JSON	TECHNICAL_STACK	STORAGE-JSON	The lightweight data-interchange format used for the flat-file database located at <code>~/todos.json</code> and for optional machine-readable output.
todo clear	BUSINESS_DOMAIN	CMD-CLEAR	The operation that purges all entries marked as finished while preserving those still pending.